

ADVANCE PROGRAMMING

LAB TASK #02

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QUESTION : use of indexer , getter setter, delegate , try catch

```
Microsoft Visual Studio Debug Console
Error: please enter a valid number.

===== MENU =====
1. Getter / Setter
2. Indexer
3. Delegate
4. Try-Catch
0. Exit
Choose an option: 4

Enter first number: 2
Enter second number: 3
Result: 0
(finally block always runs)

===== MENU =====
1. Getter / Setter
2. Indexer
3. Delegate
4. Try-Catch
0. Exit
Choose an option: 0

Goodbye!

C:\Users\hp_laptop\source\repos\ConsoleApp1\bin\Debug\net10.0\ConsoleApp1.exe (process 14408) exited with code 0 (0x0).
Press any key to close this window . . .
```

using System;

// ----- DELEGATE TYPE -----

delegate void MessageHandler(string message);

// ----- CLASS 1: Getter / Setter -----

class Person

{

```
private int age;

// auto property (simple getter/setter)
public string Name { get; set; }

// property with validation in the setter
public int Age
{
    get { return age; }
    set
    {
        if (value < 0)
            throw new ArgumentException("Age cannot be negative!");
        age = value;
    }
}

// ----- CLASS 2: Indexer -----
class SubjectList
{
    private string[] subjects = new string[3];

    public string this[int index]
    {
        get { return subjects[index]; }
        set { subjects[index] = value; }
    }
}
```

```

    }
}

// ----- CLASS 3: Delegate -----
class MessagePrinter
{
    public void ShowNormal(string msg)
    {
        Console.WriteLine("Normal : " + msg);
    }

    public void ShowUpper(string msg)
    {
        Console.WriteLine("Upper : " + msg.ToUpper());
    }

    public void Print(string msg)
    {
        MessageHandler handler = ShowNormal;
        handler += ShowUpper;    // one delegate, two methods
        handler(msg);
    }
}

// ----- CLASS 4: Try-Catch -----
class Calculator
{

```

```

public void Divide(int a, int b)
{
    try
    {
        Console.WriteLine("Result: " + (a / b));
    }
    catch (DivideByZeroException)
    {
        Console.WriteLine("Error: cannot divide by zero.");
    }
    finally
    {
        Console.WriteLine("(finally block always runs)");
    }
}
}

```

// ----- MAIN -----

```

class Program
{
    static void Main()
    {
        // create one object of each class
        Person person = new Person();
        SubjectList subjects = new SubjectList();
        MessagePrinter printer = new MessagePrinter();
        Calculator calc = new Calculator();
    }
}

```

```
string choice;
```

```
do
```

```
{
```

```
    Console.WriteLine();
```

```
    Console.WriteLine("===== MENU =====");
```

```
    Console.WriteLine("1. Getter / Setter");
```

```
    Console.WriteLine("2. Indexer");
```

```
    Console.WriteLine("3. Delegate");
```

```
    Console.WriteLine("4. Try-Catch");
```

```
    Console.WriteLine("0. Exit");
```

```
    Console.Write("Choose an option: ");
```

```
    choice = Console.ReadLine();
```

```
    Console.WriteLine();
```

```
try
```

```
{
```

```
    switch (choice)
```

```
    {
```

```
        case "1":
```

```
            Console.Write("Enter name: ");
```

```
            person.Name = Console.ReadLine();    // setter
```

```
            Console.Write("Enter age: ");
```

```
            person.Age = int.Parse(Console.ReadLine()); // setter
```

```
            Console.WriteLine("Name: " + person.Name + ", Age: " + person.Age); // getters
```

```
            break;
```

case "2":

```
    Console.Write("Enter index (0-2): ");  
    int index = int.Parse(Console.ReadLine());  
    Console.Write("Enter subject name: ");  
    subjects[index] = Console.ReadLine(); // indexer set  
    Console.WriteLine("Subject at " + index + ": " + subjects[index]); // indexer get  
    break;
```

case "3":

```
    Console.Write("Enter a message: ");  
    printer.Print(Console.ReadLine());  
    break;
```

case "4":

```
    Console.Write("Enter first number: ");  
    int a = int.Parse(Console.ReadLine());  
    Console.Write("Enter second number: ");  
    int b = int.Parse(Console.ReadLine());  
    calc.Divide(a, b);  
    break;
```

case "0":

```
    Console.WriteLine("Goodbye!");  
    break;
```

default:

```
        Console.WriteLine("Invalid choice, try again.");
        break;
    }
}
catch (FormatException)
{
    Console.WriteLine("Error: please enter a valid number.");
}
catch (IndexOutOfRangeException)
{
    Console.WriteLine("Error: index must be 0, 1 or 2.");
}
catch (ArgumentException ex)
{
    Console.WriteLine("Error: " + ex.Message);
}

} while (choice != "0");
}
```